

Lead exposure and IQ loss in children: a Bayesian and frequentist meta-analysis of recent epidemiological studies

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Notes

- Journal: [Population Health Metrics, BMC](#)
- Using [PRISMA 2020 checklist](#) for structure
- Using [raRoB tool](#) for Risk of Bias Assessment

Abstract

Background

Lead (Pb) is a pervasive environmental pollutant originating from anthropogenic and natural sources. Although concentrations have declined globally in past decades following the phase-out of leaded gasoline, lead remains present in the environment and human blood (1,2). In Europe, primary exposure sources include dietary products, drinking water, and indoor dust (3,4). The most recent Global Burden of Disease Study estimated that in 2023, 3.4 million (95% Uncertainty Interval (UI): 2.7; 4.0 million) Disability-Adjusted Life Years (DALYs) were attributable to lead exposure in the European Union (EU) (5). This highlights the continued public health significance even of low-level exposure.

Exposure to lead is associated with multiple adverse health outcomes across the human life course, including decreased kidney function (6), liver disease (7), coronary artery atherosclerosis (8), cardiovascular mortality (9), infertility in women (10), attention-deficit/hyperactivity disorder in children (11), and antisocial behavior (12). Among these endpoints, impaired cognitive development in children — manifested as reduced IQ scores — (13–16) represents a particularly critical endpoint for public health policy. Children are especially vulnerable to lead’s neurotoxic effects due to their developing nervous systems and relatively high exposure per body mass. Because lead-associated IQ decrements are irreversible and persist throughout life, they lead to substantial and enduring societal costs (17), making continued efforts to accurately characterize the exposure-response relationship important for evidence-based policy.

Contemporary blood lead levels (BLL) in the EU have declined markedly and now predominantly fall below 5 µg/dL ([HBM4EU Dashboard](#)), well below the concentrations examined in many earlier studies (<10 µg/dL). Importantly, there is no evidence for a threshold concentration below which the effects of lead on the IQ can be assumed to be negligible (18,19). There is evidence for a supralinear exposure-response relationship with steeper IQ decrements per unit increase in BLL at low concentrations compared to higher concentration ranges (16,20). These findings highlight the need for an updated quantitative synthesis that reflects both recent epidemiological evidence and the lower exposure range now predominant in European populations, incorporating recent data to inform risk assessment at currently relevant exposure levels.

Existing burden of disease assessments at national and global scales (5,17,21–23) rely predominantly on two foundational studies: a pooled analysis of seven cohort-studies by Lanphear et al. (2005) or the subsequent re-analysis by Crump et al. (2013) (16,24). While these landmark studies have substantially shaped research and policy on childhood lead exposure, methodological issues and errors in the Lanphear et al. 2005 article have been raised (24,25) and the stepwise backward-elimination regression approach used by Crump et al. (2013), which is central to their findings, has received substantial criticism (26–28). These limitations, combined with the availability of more recent epidemiological data, underscore the need not only for an updated evidence synthesis but also for methodologically robust approaches that can provide reliable exposure-response estimates for burden of disease and risk assessments.

Methodological advances in meta-analysis offer an opportunity to enhance the utility of synthesized estimates for burden of disease and risk assessment applications. While frequentist meta-analysis has been a conventional approach in environmental epidemiology, Bayesian meta-analysis provides a coherent framework for uncertainty quantification through posterior distributions that inherently incorporate both within-study and between-study variability (29). Unlike frequentist confidence intervals, which are often misinterpreted as probabilistic (30), Bayesian credible intervals directly represent the probability distribution of the parameter of interest, facilitating transparent communication of uncertainty. Furthermore, Bayesian posterior distributions can be directly propagated into probabilistic burden of disease and risk assessments, providing seamless integration with probabilistic frameworks in public health decision-making. By comparing frequentist and Bayesian approaches, we will explore how inferential framework and uncertainty quantification influence exposure-response estimates and their application.

The aim of this study is twofold: first, to synthesize and quantitatively pool available epidemiological findings on the association between blood lead levels and IQ scores in children; and second, to conduct both a frequentist and Bayesian meta-analyses, comparing how these analytical frameworks influence exposure-response estimates, uncertainty characterization, and applicability to burden of disease and risk assessment.

Methods

- ☒ Overview: systematic review and meta-analysis, comparing a frequentist to a Bayesian approach
- ☒ Inclusion & exclusion criteria, databases, search strategy, screening process
- ☒ Risk of bias analysis method
- ☒ Description & rationale of harmonization of reported effect measures
- ☒ Description of meta-analysis methodology (frequentist & Bayesian), models and software used, heterogeneity identification, prior choice
- ☒ Sensitivity analysis

Literature Review

This review adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (31). The checklist can be found in the supplementary files. The reporting of the Bayesian methods and their results was guided by the Bayesian Analysis Reporting Guidelines (BARG) (32).

We conducted a literature review to identify published studies investigating the association between postnatal (concurrent) exposure to Pb and IQ loss in children. This study was conducted

within the EU-funded project *Partnership for the Assessment of Risks from Chemicals* (*PARC*). The literature review presented here was used in multiple studies within PARC, including one on the association of (methyl)mercury and lead on IQ loss for multiple exposure timeframes. Here, we will only report results for concurrent lead exposure and IQ loss, but the figures and tables documenting the literature review do contain parts of the search for other associations and exposure timeframes.

We searched for articles published between January 2005 and July 2023. The year 2005 was chosen because, as described in the background section, burden of disease studies predominantly used a study published in 2005 (16) and we aimed at updating the evidence base with recently published studies. Also, lead concentrations have decreased significantly in the past decades and we aimed at identifying epidemiological studies conducted in populations with currently relevant exposure levels.

Search queries in Pubmed and Scopus included the following terms: substance (Pb - including Mesh terms), outcome (IQ and neurodevelopment), population (children) and results (association, ERF) (Table 1). All titles and abstracts were screened by two reviewers. Any conflicts were resolved by a third person. Full text was then assessed by one reviewer. Publications were considered relevant according to predefined PECOTS – Population (children aged 0 to 11), Exposure (biological measures of Pb from all matrices), Comparator (no exposure or lower levels of exposure), Outcome (IQ), Timing of exposure (no restriction) and Setting (no restriction) (Table 2). We excluded reviews and background articles that inventoried the literature with no new results. We also excluded references written in languages other than English.

As described above, the literature review was conducted with as the basis for multiple studies and thus had a larger scope. For this study, only studies that used a continuous, log-transformed BLL scale were included, so that effect estimates could be harmonized. The exposure timeframe for this study was restricted to concurrent, meaning that IQ and BLL measurement must be conducted at the same time. Descriptive study data was extracted using a grid based on the one proposed by NTP OHAT (33). For all included studies, effect estimates from the most-adjusted model were extracted and the set of confounders adjusted for recorded.

Risk of Bias Assessment

To assess study quality, we conducted a Risk of Bias assessment using the rapid Risk of Bias (RoB) tool with a specific focus on human observational epidemiological studies (34). Each included publication was assessed for RoB in the domains selection, exposure, outcome, confounding, follow-up, analysis, and selective reporting. In total, 15 items were evaluated for each study. The overall RoB for each individual study was then categorized by one researcher as either low, moderate or high, following two different summary metrics provided by the RoB tool (for more information, see (34)). Each publication was first assessed by one researcher. Subsequently, consensus with a large language model (LLM) was sought. This was done by uploading the list of all raRoB items along with the respective publication. Then, the LLM was prompted to assess the study, using the raRoB items and possible answer categories. If

there were any deviations in the judgement of the researcher and the LLM, the respective item was revisited by the researcher and they made the final decision.

Harmonization of Effect Estimates

Studies investigating the association between lead exposure and IQ in children have made different assumptions regarding the shape of the exposure-response relationship. While some studies assume a linear relationship (35,36), this assumption may only be valid at very low concentrations (24). Studies that tested multiple shapes of the exposure-response function have concluded that a log-linear relationship, where BLL are logarithmically transformed, provides a more appropriate model (24,37).

The general form of regression model used for a log-linear association is:

$$y = \alpha + \beta \cdot \log_b(X) + \varepsilon \quad (1)$$

Where y represents the IQ score, α is the intercept, β is the regression coefficient (slope), X denotes BLL, b is the logarithmic base, and ε is the error term.

In the literature, different logarithmic bases are used to transform BLL: the natural logarithm ($\log_e$ where $e \approx 2.71828$), base 2, and base 10. To enable pooling of the effect estimates, we harmonized all regression coefficients and their standard error (SE) to the natural logarithm scale ($\log_e$). The calculations are documented in suppl. file 1 (Screening documentation excel). In some publications, the BLL scale was left untransformed, so on a linear scale. These were excluded from the meta-analysis, as transformation of effect estimates for linear BLL to log-linear could not be justified based on visual inspection (supplement: plots?).

Meta-Analysis Methodology

In order to quantitatively synthesize the available epidemiological literature on the association of children's blood lead levels and IQ, we conducted a meta-analysis of the results, comparing a Bayesian to a frequentist approach.

In both approaches, the same effect estimates from the identified publications were included and analogous models were chosen to enable comparison between the results. We estimated pooled effect estimates for the loss of IQ points per continuous unit increase in $\log_e$ -transformed blood lead concentration ($\mu\text{g/dL}$).

R version 4.4.2 (2024-10-31) (38) was used for all computations. All computational code for this study is publicly available on [GitHub](#).

Bayesian Meta-Analysis

For the Bayesian approach, we employed a random-effects meta-analysis model with two hierarchical levels. At the first level, the observed effect size from each study $\hat{\theta}_k$ is modeled as an estimate of that study's true underlying effect θ_k , with deviation from the true effect determined by the within-study sampling error σ_k (treated as known based on reported study statistics). At the second level, these study-specific true effects are themselves assumed to vary across studies following a normal distribution, characterized by an overall pooled effect μ and between-study standard deviation (heterogeneity) τ .

$$\begin{aligned}\hat{\theta}_k &\sim \mathcal{N}(\theta_k, \sigma_k) \\ \theta_k &\sim \mathcal{N}(\mu, \tau)\end{aligned}\tag{2}$$

Where:

$\hat{\theta}_k$ = Observed effect size for study k

σ_k = Within-study sampling standard deviation for study k

θ_k = True underlying effect for study k

μ = Overall pooled effect

τ = Between-study standard deviation (heterogeneity)

The Bayesian model was fit using the R package **brms** (39) version 2.22.0. Posterior draws were sampled using the Hamiltonian Monte Carlo algorithm (40) and the No-U-Turn Sampler (NUTS) (41). We ran four chains with 4,000 iterations each, of which 2,000 iterations were discarded as warmup, resulting in 8,000 total post-warmup draws. Convergence was assessed using the Gelman-Rubin statistic ($\hat{R}$) and effective sample size (ESS). As shown in Equation 2, heterogeneity τ was modelled as a parameter directly and I^2 was calculated using the typical within-study variance that is used in the frequentist approach so that it is comparable.

Priors

In Bayesian statistics, prior knowledge or assumptions from previously published studies, domain knowledge or beliefs are updated by the newly observed data and together result in a posterior distribution. Prior distributions must be chosen for all parameters to be estimated in the analysis, which in this case are the overall pooled effect estimate μ and between-study standard deviation (heterogeneity) τ . For both parameters, weakly-informative / conservative priors were chosen so that the posterior distributions are softly being bounded to a realistic range. Because the unit scale of the included effect estimates was harmonized to IQ shift per unit increase in $\log_e$ -transformed blood lead concentration ($\mu\text{g/dL}$), the priors will be specified on the same scale.

For the overall pooled effect μ , a normally distributed prior centered at -1 with a standard deviation (SD) of 2 was chosen. This translates to the assumption that the overall pooled effect

lies between -5 and 3 with about 95% probability. The center point of the prior was chosen to be -1 because the negative effects of lead on the developing nervous system and its toxicological mechanisms of action are well established in the scientific literature (18).

Since between-study standard deviation (heterogeneity) τ must by definition be positive, a truncated normal distribution centered at 0 with a SD of 2 was specified. This translates to the assumption that τ is small, but due to its wide SD, the prior is still vague and thus weakly-informative. In summary, the following priors were specified for the Bayesian model:

$$\begin{aligned}\mu &\sim \mathcal{N}(-1, 2) \\ \tau &\sim \mathcal{N}^+(0, 2)\end{aligned}\tag{3}$$

Prior and posterior predictive simulations were conducted using the **bayesplot** package (42).

Frequentist Meta-Analysis

For the frequentist meta-analysis we employed a random-effects model, assuming the same multilevel structure as for the Bayesian model Equation 2. Parameters were estimated via the restricted maximum-likelihood (REML) estimation from the R package **metafor** version 4.8-0 (43). The I^2 statistic was calculated as a measure of heterogeneity, representing the proportion of total variation attributable to between-study heterogeneity τ^2 .

Sensitivity Analysis

We conducted two sets of sensitivity analyses. To examine the sensitivity of the results to possibly biased study estimates, we ran both the frequentist and Bayesian models again twice, excluding (1) studies with a high RoB and (2) with a high or medium RoB.

To examine the sensitivity of the results to the choice of prior distributions, we fit the Bayesian model again twice, with alternative priors for both the main effect (μ) and heterogeneity (τ). The first set of priors was more informative (more narrow) than the main priors, with $\mu \sim \mathcal{N}(-1, 1)$ and $\tau \sim \mathcal{N}^+(0, 1)$. The second set of priors was less informative (or broader) with $\mu \sim \mathcal{N}(0, 6)$ and $\tau \sim \mathcal{N}^+(0, 6)$. Notice that for the main effect, we centered the broader prior around 0, giving equal weight to the probability of a protective and detrimental effect of lead.

More informative priors (less conservative):

$$\begin{aligned}\mu &\sim \mathcal{N}(-1, 1) \\ \tau &\sim \mathcal{N}^+(0, 1)\end{aligned}\tag{4}$$

Less informative priors (more conservative):

$$\begin{aligned}\mu &\sim \mathcal{N}(0, 6) \\ \tau &\sim \mathcal{N}^+(0, 6)\end{aligned}\tag{5}$$

Results

Literature Review

As described in the methods section, the systematic review was conducted with a larger scope, because it was used as a basis for multiple studies which is why it includes another substance (mercury, Hg). The systematic search in PubMed and Scopus yielded 1,073 records for lead. After duplicate removal, 974 records were abstract and title-screened (Pb & Hg), which lead to 149 records to be screened in fulltext (Pb & Hg). After full text screening, 36 records remained for Pb. These were then assessed for their eligibility for this study. Twelve studies were excluded because they either transformed the BLL variable into categories or left it on a linear (untransformed) scale. Four studies were excluded because they exclusively assessed the effects of prenatal lead exposure. Four studies were excluded because the cohorts included in the study were already included in another study and would have been duplicate. Four additional studies were excluded because they either solely reported effects of chemical mixtures, because they used non-linear models or failed to report essential statistics. Finally, twelve studies were included in the meta-analysis. In Figure X (PRISMA), the full screening process is shown. A file containing all extracted study data as well as detailed exclusion reasons can be found in suppl. file X (systematic review process excel).

Risk of Bias

Meta Analysis Results

Sensitivity Analysis Results

- ☒ Cite studies that might appear to meet inclusion criteria, but were excluded & explain why (no citation)
- ☐ Study characteristics (table), including original and harmonized effect measures
- ☐ Flow diagram, description of (full text) screening results
- ☐ RoB results for each included study
- ☐ Forest plots
- ☐ Sensitivity analysis results

Discussion

- ☐ General interpretation of results in light of evidence in literature
- ☐ Discuss differences, similarities, strengths and weaknesses of freq vs. Bayesian approach
- ☐ Study limitations & strengths

Conclusions

- ☐ Conclusion, with reference to aim of study
- ☐ Implications of results: policy, (scientific?) practice, future research needs

List of Abbreviations

Abbreviation	Meaning
BLL	Blood lead levels
CI	Confidence interval
EBD	Environmental Burden of Disease
EU	European Union
IQ	Intelligence quotient
LLM	Large language model
REML	Restricted maximum-likelihood
RoB	Risk of Bias
SD	Standard deviation
SE	Standard error
UI	Uncertainty Interval
DALY	Disability-adjusted Life Year

Acknowledgements

The authors thank Rafiq Benchrih (Department of Epidemiology and Public Health, Sciensano, Belgium), Jurgen Buekers (Flemish Institute for Technological Research, VITO, Belgium), Carla Martins (National School of Public Health, NOVA University of Lisbon, Portugal), Emily Mc Vey (Institute for Public Health and the Environment, RIVM, The Netherlands), Anthony Purece (Flemish Institute for Technological Research, VITO, Belgium) and Sofie Thomsen (Technical University of Denmark, DTU, Denmark) for their help in the screening process.

Funding

This work was carried out in the framework of the European Partnership for the Assessment of Risks from Chemicals (PARC) and has received funding from the European Union's Horizon Europe research and innovation programme under Grant Agreement No 101057014. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the Health and Digital Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

References

1. Salamanca-Fernández E, Peinado FM, Esteban-López M, Puklova V, Poyatos RM, Govarts E, et al. [Sociodemographic determinants and temporal variability of blood lead levels \(2003–2019\) in a pooled analysis of nine studies in four European countries](#). Sci Rep. 2025 Oct;15(1):34562.
2. Lacerda D, Pestana IA, Santos Vergilio CD, de Rezende CE. [Global decrease in blood lead concentrations due to the removal of leaded gasoline](#). Chemosphere. 2023;324:138207.
3. European Food Safety Authority. [Lead dietary exposure in the European population](#). EFS2. 2012 Jul;10(7).
4. Sy M, Eleftheriadou D, Jung C, Lindtner O, Karakitsios S, Sarigiannis D, et al. [Assessment of the Long-Term Exposure to Lead in Four European Countries Using PBPK Modeling](#). Expo Health. 2024 Feb;16(1):21–39.
5. Brauer M, Roth GA, Aravkin AY, Zheng P, Abate KH, Abate YH, et al. [Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2013;2021: A systematic analysis for the Global Burden of Disease Study 2021](#). The Lancet. 2024;403(10440):2162–203.
6. Harari F, Sallsten G, Christensson A, Petkovic M, Hedblad B, Forsgard N, et al. [Blood Lead Levels and Decreased Kidney Function in a Population-Based Cohort](#). American Journal of Kidney Diseases. 2018 Sep;72(3):381–9.
7. Wan H, Wang Y, Zhang H, Zhang K, Chen Y, Chen C, et al. [Chronic lead exposure induces fatty liver disease associated with the variations of gut microbiota](#). Ecotoxicology and Environmental Safety. 2022 Mar;232:113257.

8. Rosengren E, Barregard L, Sallsten G, Fagerberg B, Engström G, Fagman E, et al. [Exposure to Lead and Coronary Artery Atherosclerosis: A Swedish Cross-Sectional Population-Based Study](#). JAHA. 2025 Mar;14(5):e037633.
9. Lanphear BP, Rauch S, Auinger P, Allen RW, Hornung RW. [Low-level lead exposure and mortality in US adults: A population-based cohort study](#). The Lancet Public Health. 2018 Apr;3(4):e177–84.
10. Chang SH, Cheng BH, Lee SL, Chuang HY, Yang CY, Sung FC, et al. [Low blood lead concentration in association with infertility in women](#). Environmental Research. 2006 Jul;101(3):380–6.
11. Dimitrov LV, Kaminski JW, Holbrook JR, Bitsko RH, Yeh M, Courtney JG, et al. [A Systematic Review and Meta-analysis of Chemical Exposures and Attention-Deficit/Hyperactivity Disorder in Children](#). Prev Sci. 2024 May;25(S2):225–48.
12. Shaffer RM, Carlson LM, Hester K, Kim H, Duffney P, Forsyth JE, et al. [Lead exposure and antisocial behavior: A systematic review of human and animal evidence](#). Environment International. 2025;109786.
13. Baghurst PA, McMichael AJ, Wigg NR, Vimpani GV, Robertson EF, Roberts RJ, et al. [Environmental Exposure to Lead and Children’s Intelligence at the Age of Seven Years: The Port Pirie Cohort Study](#). N Engl J Med. 1992 Oct;327(18):1279–84.
14. Needleman HL. [Low-Level Lead Exposure and the IQ of Children: A Meta-analysis of Modern Studies](#). JAMA. 1990 Feb;263(5):673.
15. Bellinger DC, Stiles KM, Needleman HL. [Low-Level Lead Exposure, Intelligence and Academic Achievement: A Long-term Follow-up Study](#). Pediatrics. 1992 Dec;90(6):855–61.
16. Lanphear BP, Hornung R, Khoury J, Yolton K, Baghurst P, Bellinger DC, et al. [Low-Level Environmental Lead Exposure and Children’s Intellectual Function: An International Pooled Analysis](#). Environmental Health Perspectives. 2005;113(7):894–9.
17. Larsen B, Sánchez-Triana E. [Global health burden and cost of lead exposure in children and adults: A health impact and economic modelling analysis](#). The Lancet Planetary Health. 2023;7(10):e831–40.
18. EFSA Panel on Contaminants in the Food Chain (CONTAM). [Scientific Opinion on Lead in Food](#). EFSA Journal. 2010;8(4):1570.

19. Schwartz J. [Low-Level Lead Exposure and Children's IQ: A Metaanalysis and Search for a Threshold](#). Environmental Research. 1994 Apr;65(1):42–55.
20. Canfield RL, Henderson CR, Cory-Slechta DA, Cox C, Jusko TA, Lanphear BP. [Intellectual Impairment in Children with Blood Lead Concentrations below 10 \$\mu\$ g per Deciliter](#). N Engl J Med. 2003 Apr;348(16):1517–26.
21. Carrington C, Devleeschauwer B, Gibb HJ, Bolger PM. [Global burden of intellectual disability resulting from dietary exposure to lead, 2015](#). Environmental Research. 2019;172:420–9.
22. Figueroa JL, Rodríguez-Atristain A, Bautista-Arredondo LF, Treviño CL, Martínez MR, Valdivia BT, et al. [Loss of cognitive function in Mexican children due to lead exposure and the associated economic costs](#). Environmental Research. 2024 Dec;263:120013.
23. McFarland MJ, Hauer ME, Reuben A. [Half of US population exposed to adverse lead levels in early childhood](#). Proc Natl Acad Sci USA. 2022 Mar;119(11):e2118631119.
24. Crump KS, Van Landingham C, Bowers TS, Cahoy D, Chandalia JK. [A statistical reevaluation of the data used in the Lanphear et al. \(2005 \) pooled-analysis that related low levels of blood lead to intellectual deficits in children](#). Crit Rev Toxicol. 2013;43(9):785–99.
25. Lanphear BP, Hornung R, Khoury J, Yolton K, Baghurst P, Bellinger DC, et al. [Erratum: “Low-Level Environmental Lead Exposure and Children's Intellectual Function: An International Pooled Analysis”](#). Environ Health Perspect. 2019 Sep;127(9):099001.
26. Thompson B. [Stepwise Regression and Stepwise Discriminant Analysis Need Not Apply here: A Guidelines Editorial](#). Educational and Psychological Measurement. 1995 Aug;55(4):525–34.
27. Harrell FE. [Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis](#). Cham: Springer International Publishing; 2015. (Springer Series in Statistics).
28. Smith G. [Step away from stepwise](#). J Big Data. 2018 Dec;5(1):32.
29. Grant R, Di Tanna GL. [Bayesian Meta-Analysis: A Practical Introduction](#). 1st ed. Boca Raton: Chapman and Hall/CRC; 2025.
30. Hoekstra R, Morey RD, Rouder JN, Wagenmakers EJ. [Robust misinterpretation of confidence intervals](#). Psychon Bull Rev. 2014 Oct;21(5):1157–64.

31. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. [The PRISMA 2020 statement: An updated guideline for reporting systematic reviews](#). *BMJ*. 2021 Mar;n71.
32. Kruschke JK. [Bayesian Analysis Reporting Guidelines](#). *Nat Hum Behav*. 2021 Aug;5(10):1282–91.
33. NTP OHAT. Handbook for Conducting a Literature-Based Health Assessment Using OHAT Approach for Systematic Review and Evidence Integration. 2019;
34. Plate K, Knüppel S, Hornbacher A, Greiner M, Müller-Graf C. [Development of a tool for rapid assessment of the evidence of human observational epidemiological studies with focus on risk of bias in the context of public health and risk assessment](#). *EFS3*. 2024 Jul;21(7).
35. Halabicky OM, Pinto-Martin JA, Compton P, Liu J. [Low level lead exposure in early childhood and parental education on adolescent IQ and working memory: A cohort study](#). *Journal of Exposure Science & Environmental Epidemiology*. 2023;33(2):168–76.
36. Chen A, Dietrich KN, Ware JH, Radcliffe J, Rogan WJ. [IQ and Blood Lead from 2 to 7 Years of Age: Are the Effects in Older Children the Residual of High Blood Lead Concentrations in 2-Year-Olds?](#) *Environmental Health Perspectives*. 2005;113(5):597–601.
37. Schnaas L, Rothenberg Stephen J, Flores MF, Martinez S, Hernandez C, Osorio E, et al. [Reduced Intellectual Development in Children with Prenatal Lead Exposure](#). *Environmental Health Perspectives*. 2006;114(5):791–7.
38. R Core Team. R: A Language and Environment for Statistical Computing. Vienna, Austria: R Foundation for Statistical Computing; 2024.
39. Bürkner PC. [Brms : An R Package for Bayesian Multilevel Models Using Stan](#). *J Stat Soft*. 2017;80(1).
40. Betancourt M. [A Conceptual Introduction to Hamiltonian Monte Carlo](#). arXiv; 2017.
41. Hoffman MD, Gelman A. [The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo](#). arXiv; 2011.
42. Gabry J, Mahr T. Bayesplot: Plotting for Bayesian Models. 2025.

43. Wolfgang Viechtbauer. [Conducting meta-analyses in {R} with the {metafor} package](#). Journal of Statistical Software. 2010;36(3):1–48.